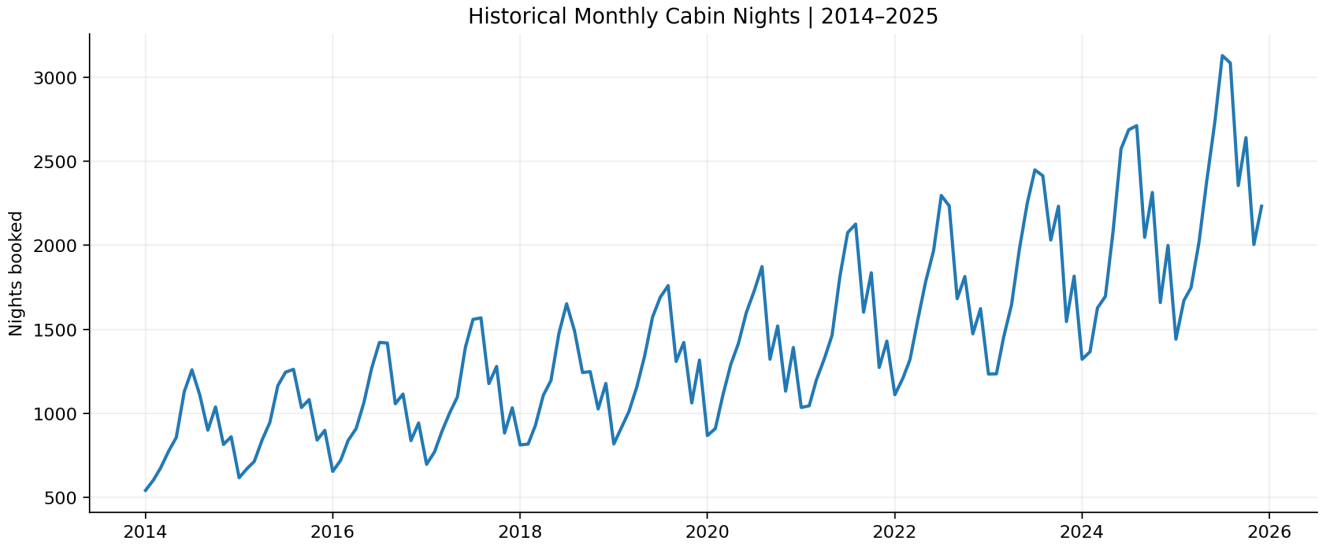
Blue Ridge Cabin Rentals

Demand Forecasting & Capacity Planning | Portfolio Case Study

Pavani Priyanka Merla | R / fpp3 coursework; visualizations reproduced using Python / statsmodels

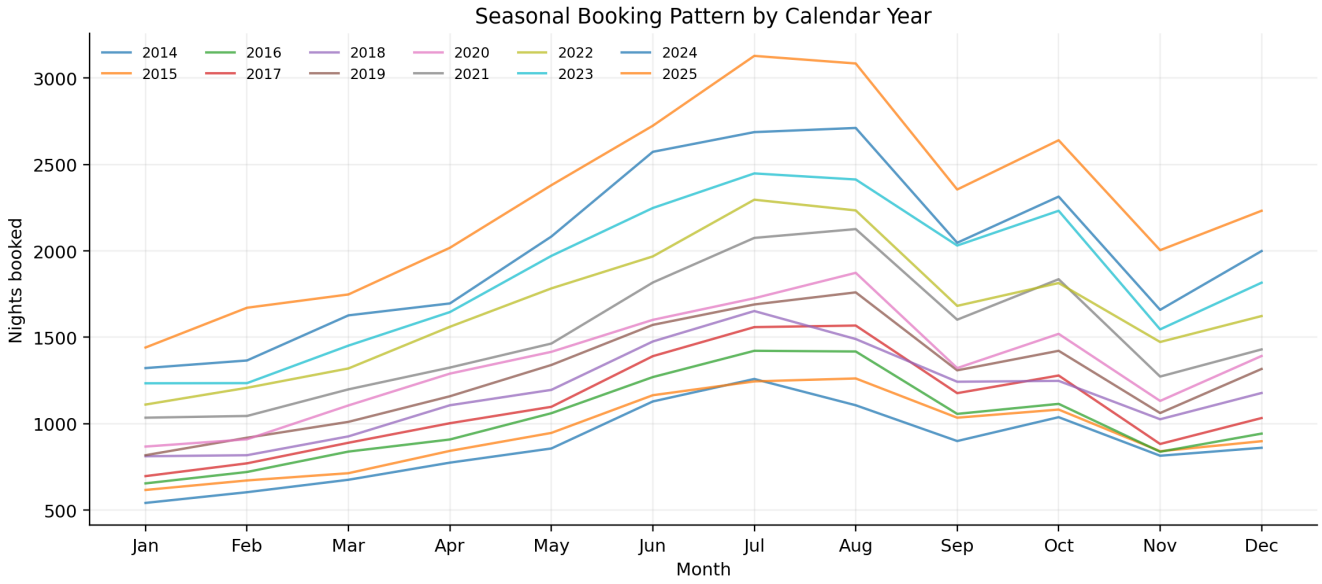
Objective. Forecast monthly cabin nights and translate seasonal demand into capacity planning decisions. The dataset contains 144 monthly observations (2014–2025), with 2024–2025 reserved for an out-of-sample assessment.

Historical demand



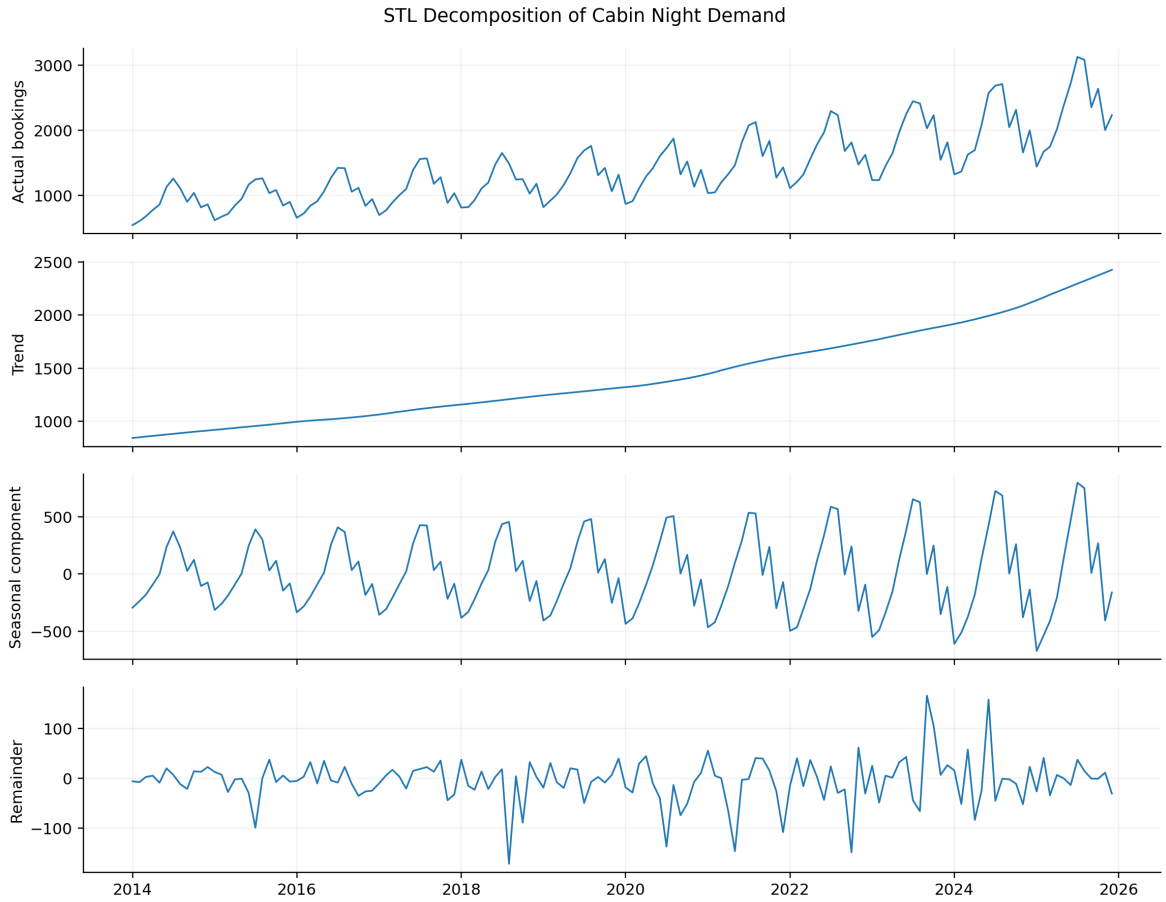
Observed bookings from the original uploaded CSV.

Demand seasonality



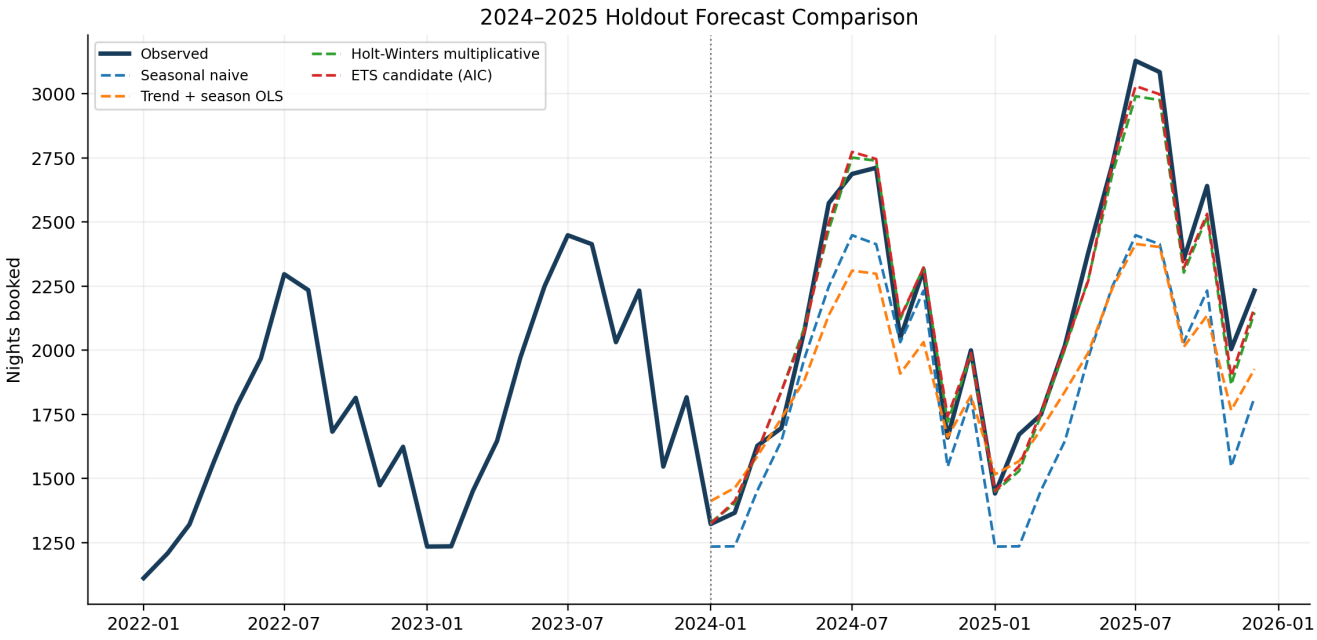
Recurring summer peaks and lower winter demand.

Trend & seasonal decomposition



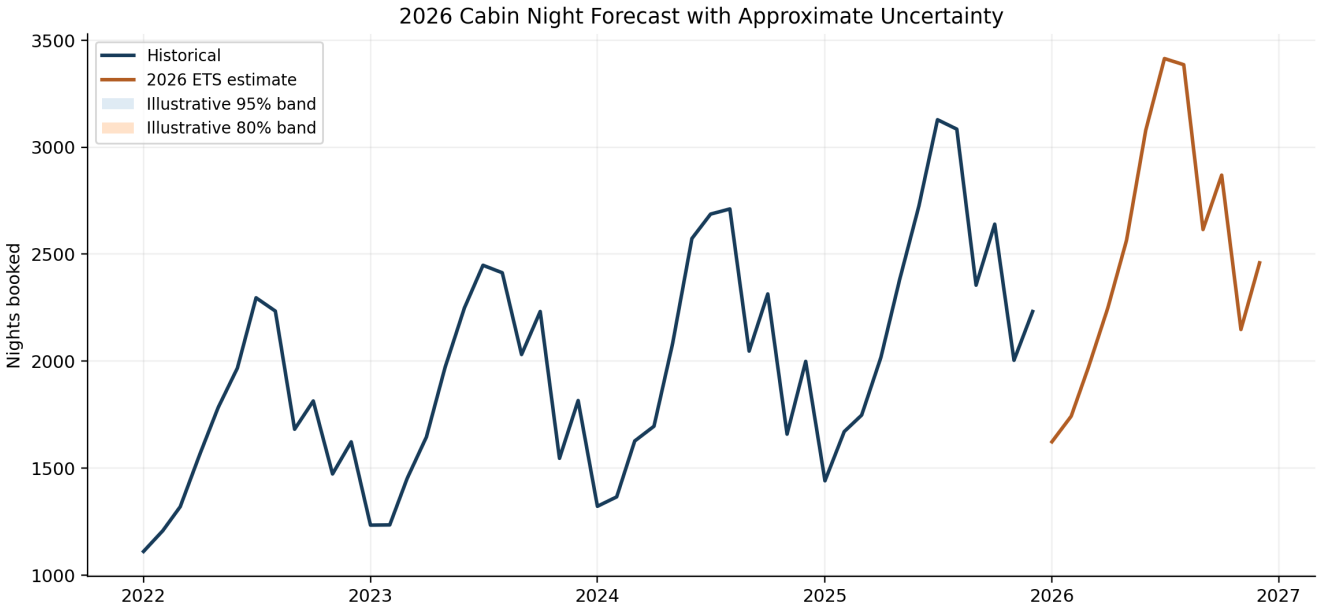
STL decomposition from actual monthly observations.

Holdout forecast comparisons



Python approximations fitted through 2023; forecasts compared with 2024–2025 actuals.

2026 forecast



Python ETS reproduction with illustrative, approximate uncertainty bands.

Methodology & outcomes

The Python holdout comparison tested seasonal naive, seasonal-trend regression, additive and multiplicative Holt-Winters, and an AIC-selected ETS candidate. Lowest holdout MASE: **ETS candidate (AIC) (0.51)**. The 2026 forecast is based on a new fit to all observed months. Original academic R code is included for reproducing its own ETS results.

Important reproducibility note: Plots 1–3 directly show uploaded data. Plots 4–5 are Python approximations and do not claim numerical equivalence with the R `fable` implementation. The forecast bands are illustrative and not validated prediction intervals. Original CSV excluded pending permission to distribute.